

Fig. 5: Hardware configuration of the 3-D gesture receiver module

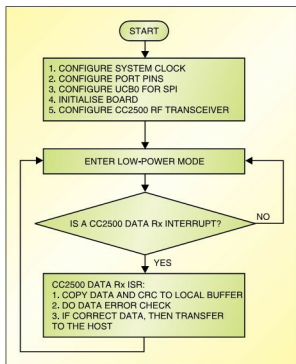


Fig. 6: Flow-chart of 3-D gesture receiver module

reception activity has been completed. It is connected to any USB2.0 port of the host PC. Communication between the host-side application and the receiver firmware is achieved using serial emulation over the USB interface. The hardware configuration is shown in Fig. 5.

The firmware flow of the 3-D gesture receiver module is shown in Fig. 6. After the RF transceiver and board-specific initial configuration is complete, the MCU waits for a reception-complete interrupt from the RF transceiver (CC2500) in a very low-power mode. When the interrupt occurs, the data is read from the hardware FIFO and transferred over USB to the host-side application.

3-D gesture and character recognition module. The received data undergoes median filtering and is sent to the recognition module. The features used for recognition consist of nondirectional unweighted graph

recognition system can be used for a range of applications. The major ones are gesture-based user interfaces, character recognition systems, human interface devices (wireless mouse, keypad, joystick, etc) and virtual reality in 3-D gaming. It has been tested for character recognition when the user gestures in the air and also for wireless mouse applications.

Further improvements

In the future, the MSP430F2274 receiver module will be replaced with C8051F34x derivatives from Silicon Laboratory to further reduce the form factor of the receiver PCB. This is because the latter has a built-in USB controller and can start working as a simple HID device, whereas currently the TUSB3410 has been used as a USB-to-UART bridge for host PC interfacing for which serial-over-USB driver is using.

A homegrown RTOS having a very small footprint (2-4 kB) and IEEE 802.15.4/ZigBee im-

plementation for RF communication is underway to achieve temporal power control in the transmitter module. TI has developed a ZigBee network processor (CC2480) that can be easily interfaced with MSP430F2274. Replacing CC2500 with CC2480 will result in reduction of the footprint for 802.15.4 and ZigBee stack. Usually, a 802.15.4 stack takes 25-30 kB, whereas ZigBee takes 60-70 kB of flash, which will be a constraint for current MSP430F2274 implementation. ●

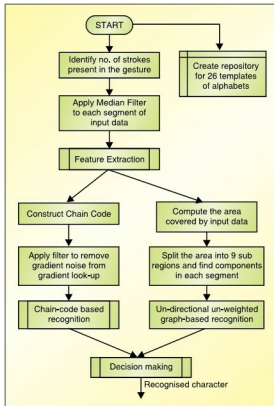


Fig. 7: Flow-chart of recognition algorithm

based detection for straight-line gestures and characters. Chain-code-based detection is used for curved gestures. Each method recognises the input with some confidence factor lying in the closed interval (0,1). A multifactorial approach is used to make final decision about the recognised gestures and characters.

Fig. 7 shows the flow-chart of the recognition algorithm. A recognition accuracy of more than 90 per cent has been achieved with input test vectors created by 22 users, where each user is asked to write each letter five times.

Host application. This 3-D gesture

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